BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 010 · CPU Architecture, Registers and Fetch–Execute Cycle

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which technology does a USB flash drive use?

Q2. Why compress a file before sending it?
A. To encrypt its contents
B. To send less data
C. To add more pixels
D. To increase its size

Q3. What does lossy compression do?

Q4. A program is compressed before being sent. “Lossless compression removes source-code characters.” Correct this. Why does it matter here?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

MAR — Stores the address of a memory location being accessed.

MDR — Stores data/instructions being transferred.

PC — Stores address of the next instruction.

ACC — Stores results of calculations.

New learning

CPU roles: The Control Unit coordinates operations and decodes instructions; the ALU performs arithmetic and logical operations.

Registers: PC: next instruction address; MAR: address being accessed; MDR: transferred data/instruction; ACC: calculation result.

Kitchen memory aid: PC = recipe bookmark; MAR = shelf location; MDR = ingredient fetched; ACC = mixing bowl for results.

Instruction cycle: Fetch the instruction, decode it and execute it; then repeat. Cache keeps frequently used data and instructions close to the CPU.

Worked example

Explain a fetch using the PC, MAR, MDR and the Control Unit.

1. The PC identifies the address of the next instruction.

2. The address is copied to MAR.

3. The requested instruction transfers from memory into MDR.

4. The PC updates to reference the next instruction in sequence.

5. The Control Unit decodes the fetched instruction; it can then be executed, possibly using the ALU.

We Do · Practise with guidance

Guided 1

Which register stores the next instruction's address?

Worked method: Program Counter (PC).

Guided 2

Where is the address of the memory location being accessed stored?

Worked method: Memory Address Register (MAR).

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BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Solid-state storage with no moving read/write mechanism.

Q2. B — To send less data

Q3. It discards some information, so the exact original cannot necessarily be restored.

Q4. Lossless compression restores every character exactly. Losing characters could break the program. A thumbnail can lose some detail; source code cannot.

I Do / We Do pages · Guided practice

Q1. Program Counter (PC).

Q2. Memory Address Register (MAR).

Q3. ALU performs arithmetic; accumulator can hold the result.

You Do pages · Code First

```
a = 7
b = 5
print(a + b)
```

Expected result: 12

You Do pages · Independent questions

Q1. MAR: accessed address [1]; MDR: data or instruction being transferred [1].

Q2. PC stores address of next instruction [1], providing the location to fetch next [1].

Q3. CU decodes/co-ordinates [1], controls the operation [1], ALU calculates [1], ACC may store the result [1].

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Guided 3

Which unit calculates a sum and where might the result be held?

Worked method: ALU performs arithmetic; accumulator can hold the result.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the number printed.

```
a = 7
b = 5
print(a + b)
```

predict

Question 1

State what MAR and MDR each store. [2 marks]

Question 2

Explain the role of the Program Counter when fetching an instruction. [2 marks]

Question 3

Explain how the Control Unit, ALU and accumulator contribute to processing an addition instruction. [4 marks]

Exit Ticket · Answer before checking

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BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

MDR stores the address of the next instruction.